MILITARY SPECIFICATION

VEHICLE, COMBAT, ANTI-TANK, IMPROVED TOW VEHICLE (ITV) (LESS TOW WEAPON), M901

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the integrated characteristics of one type of armored full-tracked vehicle designed as an anti-tank guided missile carrier for cross country and amphibious operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

BB-N-411	- Nitrogen, Technical.
TT-E-529	- Enamel, Alkyd, Semigloss.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-514	- Plates: Identification, and Marking, Blank.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-H-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-W-13855	- Weapon, Small Arms and Aircraft Armament Subsystems, General Specification for.
MIL-P-27401	- Propellant Pressurizing Agent, Nitrogen.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-C-62310	- Carriers, Armored, Full-tracked: M113A2, M106A2, M125A2 and M577A2, and Chassis, Weapon, Full-tracked: M741A1, and M901.
MIL-T-62327	- TOW Vehicle: Combat, Anti-tank (less Tow Weapon) M901, Processing for Storage and Shipment of.

STANDARDS

FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-193	- Painting Procedures, Tactical Vehicles (Tracked and Wheeled).
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-1246	- Product Cleanliness Levels and Contamination Control Program.
MIL-STD-1275	- Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles.
MIL-STD-1474	- Noise Limits for Army Materiel.

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2.1.2 Other Government documents, and Government drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

PURCHASE DESCRIPTIONS

ATPD-l	- Kit, Improved TOW Vehicle (ITV) for Combat Vehicle Anti-tank, Improved TOW Vehicle, M90l.
MIS-14061	- Missile Interim Specification Launcher, Tubular Guided Missile, M220.

DRAWINGS

ARMY

8736977	- Combat Vehicle, Anti-tank, Improved TOW Vehicle, M90l.
10942803	- Lubrication Chart, Production, Fuel and Coolant.
12277600	- Target Board.
12277608	- Pinhole Cover.
12277611	- Boresight Adapter.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article (preproduction model). A first article of the vehicle to be furnished under this specification shall be produced after award of contract and before beginning production. First article shall be representative of units to be manufactured in quantity, and shall be constructed with tools and equipment normally used in quantity production (see 6.3).

3.2 Initial production vehicles. Vehicles manufactured on initiation of quantity production shall be inspected, and shall pass initial production inspection (see 4.4). Initial production vehicles shall be representative of vehicles to be furnished under this specification, and shall include all current modifications. No modifications shall be applied to subsequent vehicles without prior approval of the Government.

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3.2.1 Improved TOW Vehicle (ITV) kit reliability. The ITV kit shall demonstrate a minimum of 180 hours Mean-Time-Between-Failure (MTBF) when installed and operated in an M113A1 vehicle in accordance with the operational profile (see 3.2.3). For calculating MTBF, the definition of combat mission failures in the Improved TOW Vehicle RAM-D Failure Definition and Scoring Criteria shall be used (see 6.9).

3.2.2 ITV kit maintainability. The ITV kit, when installed and operated in an M113A2 vehicle, shall not exceed the following maintenance ratios:

Organizational maintenance:	0.05
Direct support maintenance:	0.04
Combined maintenance:	0.09

The maintenance ratio shall be calculated by dividing the total maintenance man-hours (exclusive of daily crew checks and services) by the total operating hours.

3.2.3 Operational profile.

3.2.3.1 Chassis. Apportioned test mileage:

a. 20 percent hard-surfaced roads.
b. 40 percent secondary roads.
c. 40 percent cross-country.

3.2.3.2 ITV kit. The operational profile for the ITV kit shall consist of:

- 11.5 hours duration
  - 1.9 hours active weapon station hours
- 10 missile firings of simulated missile firings
- 7 reload cycles
- 7 erect/traverse/stow cycles
- 108 tracking cycles of 60 degrees each
- 25 miles of travel as follows:
  - 5 miles primary road
  - 10 miles secondary road
  - 9 miles cross-country
  - 1 mile cross-country, 5 mph or less, with launcher erect

3.3 Materials. Materials shall be as specified herein and in referenced specifications, standards and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.8).

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3.3.1 Qualified products. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPLs) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of supplier and part or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document and name of supplier(s).

3.3.2 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.4 Construction. The M90l kit shall be integrated with the M90l chassis to form a complete vehicle in accordance with drawing 8736977 and applicable specifications and standards. Riveting, welding practices and quality shall be the same on each vehicle. A completed vehicle shall meet all requirements specified herein, and be of a quality that assures a vehicle free of defects that would compromise, limit, or reduce its performance.

3.4.1 M901 kit. The M901 kit, when integrated with the M901 chassis shall conform to ATPD-l in its entirety.

3.4.2 M90l chassis. The M90l chassis, shall conform to MIL-C-623l0.

3.4.3 TOW components interface. All design functions and capabilities of the TOW Missile Guidance Set (MGS), daysight, nightsight and missile as specified in MIS 14061 shall be retained in their M901 system utilization. Functional interfaces with these standard TOW system components shall be as shown on their applicable drawings.

3.4.4 Seals.

3.4.4.1 Weather seals. Weather sealed cable routings and all other ITV components shall restrict entrance of water to a maximum of 10 drops per minute, collectively, into the vehicle. Other foreign matter shall be prevented from entering the vehicle.

3.4.4.2 Bearing seals. When fording, operating amphibiously, or in mud, sand or snow, seals shall prevent entrance of foreign matter into the bearings, which are exposed to contamination during these operations. Bearing seals shall restrict the leaking of lubricants from bearings.

3.4.5 Hatches and vision devices. All hatches and vision devices furnished shall conform to applicable drawings. The gunner's and driver's hatches shall provide three positions; closed, pop-up and fully open. The vehicle shall contain provisions for latching the cargo hatch in a 1/4 open position and 1/2 open position in addition to the fully closed and fully open cargo hatch

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provisions. Positive locks shall be provided for all hatches in all required positions. Vision device receptacles incorporated in the vehicle shall accept Government supplied vision devices without binding or interference, and provide adequate support during all vehicle operating conditions. All closed hatches and installed vision devices shall restrict entrance of water to a maximum of 10 drops per minute, collectively, into the vehicle.

3.4.6 Electrical systems.

3.4.6.1 Electrical circuits. The electrical system of the weapon station shall be compatible with a vehicular system power supply which conforms to the requirements of MIL-STD-1275, except for surge and spike conditions. It shall enable the weapon station to successfully perform all of its specified design functions when furnished power from the batteries and operational electric power system of the vehicle, or emergency power from a single TOW MGS battery. Communication and electrical circuits shall maintain continuity from end to end, without any shorts from one pin to the other, or from each pin to the connector shell, to assure proper function of communications and other electrical equipment.

3.4.6.2 Generating system voltage (alternator). Alternator shall produce 50 $\pm$ 5 amperes at 4,000 $\pm$ 100 rpm, and the voltage regulator rheostat shall be adjustable to deliver 28 $\pm$ 0.5 volts at the chassis batteries.

3.4.6.3 Lights. The lights, as specified on applicable drawings, shall operate throughout all vehicle operating conditions. Electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.4.7 Adjustment mechanisms. All adjustment mechanisms, including gunner's and squad leader’s seats, shall function properly and maintain adjustment settings during all vehicle operating conditions.

3.4.8 Welding repairs. Welding repairs of any kind shall be made only when, and to the extent, specifically authorized by the applicable drawings, specifications or welding procedures which have been approved by the procuring activity.

3.4.9 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction throughout all ranges of operation under all operating conditions.

3.4.10 Hydraulic lines. Hydraulic lines and connections shall be internally clean prior to initial filling of the system with hydraulic fluid conforming to MIL-H-5606 or MIL-H-6083. Leakage at the hydraulic lines, connections and reservoirs shall not exceed a “seep” condition (see 6.6).

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3.4.11 Weight. The M901 vehicle weight, when combat loaded in accordance with applicable drawings, shall not exceed 26,300 pounds.

3.4.12 Stowed equipment. All interior and exterior Basic Issue Items (BII), shall be stowed (combat loaded) on the vehicle in the spaces provided. Items shall fit properly and shall not interfere with the operation of vehicle and components in any manner.

3.4.12.1 Stowage list, hard equipment items. Manufacturer and depot installed hard equipment shall consist of:

a. M60 Machine gun (or simulator).
b. Radio and associated equipment.
c. Tow equipment.
d. Simulated TOW missile round.
e. Ammunition boxes.
f. Water cans.

3.4.13 Fastener torque. All fastener torque shall conform to good commercial practice. Torque values shall be appropriate for size of bolt and fastener attaching conditions.

3.5 Break-in run. Prior to tests, each vehicle shall be given a break-in run for a distance of fifty miles or more on smooth, level, hard-surfaced roads, except that a section of road shall be as shown in figure 1. Figure 1 section shall be traversed at least once every five miles. After the break-in run, there shall be no indication of damaged components, loss of original adjustment, or other defects that may cause faulty vehicle operation subsequent to Government acceptance. The vehicle shall also be subjected to the slope test (see 4.7.20).

3.5.1 Preparation. Prior to break-in run, except for equipment and component installation and servicing performed by the contractor as a final processing of the vehicle for acceptance, the vehicle prepared for system integration shall be completely assembled with all adjustments completed. The vehicle, including all components requiring lubrication, shall be lubricated and serviced as specified in drawing 10942803. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified. The vehicle engine air intake filter shall be inspected to assure that it is free of chips and dirt.

3.6 Performance. A complete vehicle, loaded to 26,300 pounds and serviced as specified in 3.5.1, shall perform as specified herein with launcher in stowed position unless otherwise noted. Vehicle, serviced and equipped for prevailing climatic conditions, shall operate as specified with

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or without special equipment. Performance shall be demonstrated on smooth, level, hard-surfaced roads free of loose material, except a section of road shall be as shown in figure 1.

3.6.1 Environmental.

3.6.1.1 Climatic conditions. The vehicle shall operate in ambient air temperatures of 125°F to minus 25°F.

3.6.1.2 Shock and vibration. The vehicle, with the weapon station launcher in the stow position, shall not be damaged, nor shall its performance be impaired, after exposure to shock and vibration experienced during the operating conditions specified herein. The vehicle, with the weapon station launcher in the erect position, shall not be damaged, nor shall its performance be impaired after exposure to the shock and vibration environment experienced during the launcher erect condition defined in the operation profile. The vehicle shall not damage nor impair performance of the TOW missile or TOW system components beyond that which would result from exposure to the normal shock and vibration environment experienced during operating conditions specified herein.

3.6.2 Bilge pumps. The bilge pumps, installed in accordance with applicable drawings, shall remove water from the vehicle at a minimum rate of 44 gallons per minute (gpm) each, with a discharge head of 4 1/2 feet, and with pump motor operating on 27.5 volts impressed at the motor.

3.6.3 Speed. The vehicle shall sustain speeds up to 32 mph without damage to power plant and suspension system. Performance shall be demonstrated on smooth, level, hard-surfaced road. With vehicle traveling between 15 mph and maximum speed, drift shall not exceed three feet in 100 feet.

3.6.4 Acceleration. With the vehicle at a standstill, the transmission in 1-3 range and engine at idle speed, the vehicle shall accelerate from 0 to 20 mph in not more than 13 seconds on smooth, level, hard-surfaced road.

3.6.5 Braking.

3.6.5.1 Stopping and vehicle control. The vehicle shall stop within a distance of 57 feet from point of service brake application when traveling at a speed of 20 mph on smooth, level, hard-surfaced roads free of loose material. Within the stopping distance, the drift shall not exceed three feet. During braking operation, steering action by the driver is permitted to minimize drift.

3.6.5.2 Holding. The vehicle shall hold stationary on a 60 percent longitudinal grade with brakes locked for parking and transmission in neutral.

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3.6.6 Water crossing.

3.6.6.1 Shallow water crossing. The vehicle shall ford shallow, hard-bottomed, salt or fresh water channels up to the depth required to float vehicle. Water accumulated in the hull shall not exceed 1 quart per minute.

3.6.6.2 Amphibious operation. The vehicle, launched amphibiously, shall negotiate water of depth sufficient to float vehicle with height of waves not exceeding one foot. When afloat in calm water not less than 15 feet in depth, a speed of not less than three mph shall be attained. The vehicle shall operate within a 30 foot corridor when crossing calm, open water. Water accumulation in the hull shall not exceed 1 quart per minute.

3.6.7 Trench crossing. The vehicle shall cross trenches up to 36 inches in depth, and up to 66 inches in width, without damage to the vehicle.

3.6.8 Vertical obstacles. The vehicle, operating at low speeds, shall cross vertical obstacles up to 24 inches high in forward gear, and up to 14 inches high in reverse gear, without damage to the vehicle.

3.6.9 Climbing. The vehicle shall ascend a 60 percent grade in forward and reverse gear from a standstill position on the grade without stalling or damaging the power plant or power train.

3.6.10 Grades and slopes. The vehicle, with weapon station, shall be operable on 60 percent forward and reverse grades and on 30 percent side slopes. Side slope limits for missile firing are $\pm$ 10 degrees as imposed on the TOW missile system. The weapon station brakes, when engaged, shall prevent any azimuth rotation under all specified slope conditions.

3.6.11 Ramp system. Ramp raising time, from fully lowered position to closed position, shall be not more than 18 seconds at a maximum of 1200 rpm of the engine.

3.6.12 Launcher. Capability shall be provided for driving the launcher and erection arm assembly to any of three allowable positions, stow, erect or load. The allowable position changes shall be:

a. Stow to erect.
b. Erect to stow.
c. Erect to load.
d. Load to erect.

3.6.12.1 Mission functions. Target detection, acquisition, tracking, slewing and missile arming, and firing shall be the primary operational functions performed with the erect launcher.

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3.6.12.1.1 Power. Operation of the erection system in any mode shall be prohibited in the absence of both primary and emergency power. The erection system shall be driven by the erection drive motor when primary power is available. The erection system shall be driven between stow and erect with the hand crank drive on the erection arm, when emergency power is used as specified in 3.6.16.1.

3.6.12.2 Erect position. When the erect mode is selected and the launcher and erection arm are driven to the erect position as specified in 3.6.12, the erection arm and the elevation drive assembly shall be secured in the erect position by the extension of lock cylinders. When the weapon station power (primary or emergency) is on, and hydraulic accumulator indicator pressure is 1200 psi, the erection lock cylinders shall engage.

3.6.12.2.1 Erection lock cylinders. The lower erection lock cylinders shall remain engaged, securing the erection arm in the erect position, when hydraulic system pressure has first been bled to the precharge level and then primary and emergency power to the weapon station removed. The erection lock cylinders shall always retract, when load or stow position is selected and hydraulic system pressure is 1200 psi or greater.

3.6.12.2.2 Erection motor speed and time. The erection motor speed shall be controlled, so as to limit acceleration and shock loading of the launcher. The motor shall stop, without hunting or overshooting, when the erection arm reaches the desired position. The time required to erect and lock the launcher from the stow position shall not exceed eight seconds.

3.6.12.3 Stow mode. The stow mode shall be provided as the normal travel position. The launcher and erection arm assemblies shall rest on the stow vertical supports, and within the lateral restraints, during all vehicle operating conditions without damage to the vehicle.

3.6.12.4 Load mode. When in the load mode the launcher shall be at an azimuth angle of 0 ± 1 degree, the elevation angle shall be at the maximum displacement, and the erection arm shall be at an angle which is 63 ± 5 degrees from the erect position of the erection arm. This position shall be maintained in the load mode.

3.6.12.5 Azimuth and elevation positional capability. Capability shall be provided to drive the vehicle weapon station continuously, in either azimuth direction, using the hand controller, and to drive the launcher to any elevation angle between 30 ± 1 degrees depression and not less than 34 degrees elevation.

3.6.12.6 Azimuth and elevation drives. The azimuth drive shall rotate the weapon station about the weapon station azimuth axis. The elevation drive shall rotate the launcher assembly about the elevation axis. Control of azimuth and elevation drives shall be provided in the erect

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mode. Unless otherwise specified (see 6.2), the tracking and slewing rate performance specified herein is contingent upon regulated hydraulic pressure (3.6.12.7.3) being supplied to the azimuth and elevation drive systems.

3.6.12.6.1 Azimuth drive. The azimuth drive shall provide weapon station rate control in two speed ranges, track and slew as defined herein.

3.6.12.6.1.1 Slew speed range. At the maximum control handle displacement in traverse, the weapon station speed in the slew mode over a level arc shall be not less than 35 degrees/second.

3.6.12.6.1.2 Track speed range. The azimuth drive shall maintain a minimum rate of .1 MRAD per second.

3.6.12.6.2 Elevation rate. At the maximum control handle displacement, the launcher rate in either elevation or depression shall be not less than 12 degrees per second within 25 degree depression to 30 degree elevation. The elevation drive system shall provide a capability for maintaining a minimum elevation rate of .1MRAD per second.

3.6.12.6.3 Target tracking. The weapon station, when operated by qualified personnel shall remain on a crossing target 85 percent of the time during the tracking event. The time duration of a tracking event shall be a minimum of 8 seconds. The target shall provide the following:

a. Square, 4.25 MRAD per side.
b. Range of peak azimuth rates, 0. to 25. MRAD per second.
c. Range of peak elevation rates, 0. to 2. MRAD per second.

3.6.12.7 Hydraulic system.

3.6.12.7.1 Power supply. The hydraulic system power supply shall be an electric pump motor. Primary power to the weapon station shall be required for its operation. Normal system accumulator pressure range shall be 1525 psi to 1900 psi, except when overridden by the erection drive. Maximum accumulator pressure shall be limited to 2350 psi. Pressure loss shall not exceed 300 psi in 7 minutes for non-active, erect launcher and fully pressurized system.

3.6.12.7.2 Fluid. The hydraulic system shall use fluid in accordance with MIL-H-5606 or MIL-H-6083. The hydraulic system reservoir dipstick shall indicate the full fluid level. The hydraulic system shall conform to cleanliness level 300 as specified in MIL-STD-1246.

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3.6.12.7.3 Pressurization. The accumulator shall be pressurized to the required precharge using nitrogen gas specified by MIL-P-27401, type I, grades A, B, or C, or BB-B-411, type I, grades A or B. The accumulator nitrogen precharge shall be 1000 psi ± 100 psi (at 72°F, with zero hydraulic pressure on the system). The azimuth and elevation drive systems shall be supplied with regulated hydraulic pressure within the range of 1400 to 1500 psi. A manual hydraulic pump shall be provided for the manual backup mode, as specified in 3.6.16.

3.6.12.8 Safety interrupt and override.

3.6.12.8.1 Safety interrupt erected launcher. Safety interlocks shall be provided to protect crew and equipment when operating the weapon station and firing the TOW missile. An override capability shall be provided which will permit target tracking and missile firing in the event of operational necessity. The interrupt conditions and resulting functions which shall be allowed are summarized in table I, together with override provisions indicated. The first interrupt azimuth/elevation zones shall be as shown in table II. If the gunner's hatch is open, the erection drive is inhibited from erect to load and may not be overridden, and the azimuth and elevation drive action is inhibited but may be overridden.

3.6.12.8.2 Stowed launcher. The erection drive system shall not be inhibited when the launcher is in the stow position. The first interrupt override shall be inhibited in the stow position.

3.6.12.8.3 Load positioned launcher. The erection drive system shall not be inhibited when the launcher is in the load position. The fire interrupt override shall be inhibited in the load position.

TABLE I. M901 Safety interrupt conditions (launcher erect, on primary power).

Condition	Arm	Fire	Stow	Load	Track/slew
Driver's hatch open	Yes	No 1/	Yes	Yes	No 1/
Gunner's hatch open	Yes	No 1/	Yes	No	No 1/
Cargo hatch open (intermediate detent position)	Yes	No 1/	No	Yes	Yes
Cargo hatch fully open	Yes	No 1/	Yes	Yes	Yes
Launcher AZ/EL in no-fire zone	Yes	No 1/	N/A	N/A	Yes

1/ With override engaged, this function is allowed.

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TABLE II. Launch AZ/EL no-fire zones.

Weapon station azimuth relative to vehicle (degree) ± 3.5 degrees	Launcher elevation angle at interrupt (degree) (equal to or less than)
347.5 – 12.5	Maximum depression – No interrupt
12.5 – 58	-25. to -22
58. – 118	Maximum depression – No interrupt
118. – 233	-16. to -13.
233. – 310	Maximum depression – No interrupt
310. – 347.5	-25. to -22.

3.6.12.9 Controls and indicators. The controls and displays for operation of the weapon station shall be grouped on the control panel, the hand control unit and at other locations in the turret, providing ready access to the gunner.

3.6.12.9.1 Control panel controls and indicators. The control panel shall contain controls, self test features, guards and indicators. The indicators shall identify selection and activation of weapon station functions, and shall display performance status.

3.6.12.9.2 Hand controller. The weapon station hand controller shall enable gunner to control weapon station azimuth, missile launcher elevation, missile firing and optical sight selection.

3.6.12.9.2.1 Dead zone. The minimum control handle rotation from the neutral position to produce turret motion in either right/left traverse, or in elevation/depression, shall not exceed 4 degrees. The control handle shall return to the neutral position when released at any position and shall not produce any drift in azimuth/elevation.

3.6.12.9.2.2 Azimuth control. Movement of the control handles clockwise/counterclockwise shall disclose no indication of binding. Full rotation about the neutral position shall be 35 ± 5 degrees in either direction.

3.6.12.9.2.3 Elevation control. Movement of the control handles in elevation/depression shall disclose no indication of binding. Full rotation about the neutral position shall be 30 ± 5 degrees in each direction.

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3.6.12.9.2.4 Hand controller indicator. A missile trigger actuated indication shall be provided to the operator. It shall indicate a closed trigger circuit, when the trigger circuit is closed and the TOW Missile Guidance Set (MGS) is installed and sufficiently powered.

3.6.12.9.3 Nightsight remote controls. Remote nightsight controls shall be provided to adjust nightsight image brightness, contrast, focus and field of view (FOV) selection. These controls shall be smooth and continuous in operation, excluding the FOV control which is two position.

3.6.12.9.4 Other controls and indicators. Controls and indicators shall also be provided to control or indicate:

a. Accumulator nitrogen pressure.
b. Weapon station roll angle (-10 to +10 degrees) with illumination.
c. Weapon station azimuth relative to the vehicle with illumination.
d. Ready to load condition. Indicator shall be illuminated and visible to personnel behind the gunner, and shall provide brightness control/lamp test features.
e. Vehicle lateral cant angle (-20 to +20 degrees) with illumination.
f. An audible alarm which sounds when the system is in the fire interrupt condition, and either remote arming device is in the armed state.
g. Cumulative time that power has been applied to the weapon station, primary or emergency.

3.6.13 Optics. The optics system shall consist of; weapon station Image Transfer Assembly (ITA) and vision blocks, squad leader's periscope (SLP), and vehicle hull periscopes/vision blocks.

3.6.13.1 Image Transfer Assembly (ITA). The ITA is a unity (1X) periscope, which serves as an optical relay between the gunner and the TOW daysight/tracker and nightsight, and also provides a wide field acquisition sight. The ITA shall provide features required to accommodate these three selectable optical input channels. It shall provide articulation, when the weapon station is positioned to the erect, stow and load positions of the weapon station launcher.

3.6.13.1.1 Sight selection. The ITA shall provide for gunner selection of an image at the gunner eyepiece from:

a. The acquisition sight, 2.8X magnification with 25 degree field of view.
b. The TOW daysight, 13X magnification with 5.5 degree field of view.
c. The TOW nightsight, 4X and 12X magnification with attending field of view either 3.4 X 6.8 degrees or 1.1 X 2.2 degrees.

The ITA shall require no more than 2 seconds to switch from one channel to any other channel.

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3.6.13.1.2 Internal atmosphere (ITA). The ITA, after being pressure sealed, shall exhibit no evidence of moisture, internal condensation, or internal image fogging. The pressurized sections of the ITA shall be initially purged and charged with nitrogen, conforming to BB-N-411, type I, class 1, grade A or B.

3.6.13.1.3 Image movement. The image transmitted to the gunner through the acquisition sight, daysight, or nightsight, shall not exhibit any random movement during weapon station operations which impair or degrade gunner vision or system tracking performance.

3.6.13.2 Squad leader periscope. The squad leader periscope (SLP) shall provide the following articulation of the line-of-sight:

Azimuth	± 360 degrees
Elevation	10 degrees
Depression	20 degrees

The SLP magnification shall be 4.0X and the field of view shall be 12.5 degrees.

3.6.14 Boresight.

3.6.14.1 Daysight/tracker. The IR line of sight of the missile tracker shall be the reference to which the TOW daysight/tracker visual line of sight is aligned. The daysight visual line of sight shall be used as a reference for alignment of the nightsight and missile launch tubes.

3.6.14.2 Nightsight. The nightsight optical line of sight shall be aligned to that of the daysight by means of the nightsight boresight collimator.

3.6.14.3 Launch tubes. The centerline of the missile launch tubes shall be aligned to the visual line of sight (LOS) of the daysight/tracker in accordance with the following requirements:

a. Left launch tube:
   Centerline pitch above LOS: 0.60 + .15, - .1 degree
   Centerline toe-in toward LOS: 0.78 ± .15 degrees

b. Right launch tube:
   Centerline pitch above LOS: 0.60 + .15, - .1 degree
   Centerline toe-in toward LOS: 0.43 ± .15 degrees

The launch tubes' boresight shall remain within the specified tolerances for the full range of launcher elevation angles.

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3.6.15 Electrical system.

3.6.15.1 Intercommunications. Five C2298/VRC intercom control boxes shall be provided as part of the vehicle. The weapon station shall provide a connection which transmits all the functions of a C2298/VRC intercom control box.

3.6.15.1.1 Weapon station intercom noise. The noise level in the weapon station intercom when the hydraulic pump motor has been induced to run shall be measured, across a 10.5K ohm resistor connected across the audio output, to be less than 350 millivolts AC/rms. The following conditions must be followed to obtain this measurement:

a. Turn the AM/1780 audio AMP power circuit breaker to ON (NOTE: Power on indicator lights). Set the MAIN POWER selection switch to INT. ONLY. Set the INT accent to OFF. Set the radio transmission selection switch to CDR + CREW.
b. Set the volume controls to full volume (full CW) and the set selector switch to INT ONLY on all C2298/VRC intercommunications controls; and
c. Connect a 150 ohm load across the MIC input of the A1A3J2 headset connector on the turret.

3.6.15.2 Slip ring circuits. The weapon station slip ring shall provide uninterrupted continuity of communications, power and signal circuits throughout continuous 360 degrees of traverse. Each slip ring power circuit shall conduct 200 amperes of constant current between the chassis and the weapon station. Each slip ring signal circuit shall conduct a maximum of 7 amperes continuous current. The operation of the slip rings shall not introduce noise or static into the communications system, which would impair intercommunication intelligibility.

3.6.15.3 Missile arming. The missile arming system shall provide remote selection/arm capability at the gunner's station.

3.6.15.3.1 Missile select. The weapon station shall provide for missile selection, which activates the arming circuits of the selected missile. An indication shall be provided which verifies that the selected missile arming circuits have been activated.

3.6.15.3.2 Missile arm/safe. A protected arm/safe selector shall be provided which serves to arm a selected missile by activating the appropriate remote arming device (RAD). The RAD shall then extend its HI-G shear pin plunger and firing circuit umbilical plug. An armed indication shall be provided to the gunner, which shows when a RAD has been activated. The safe position of the arm/safe selector shall:

a. Enable the wire-cut signal to pass from the MGS to a spent round; and
b. Energize the selected RAD to retract its HI-G shear pin plunger and firing circuit umbilical plug.

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3.6.15.3.3 Safety provisions. The missile arming system shall contain the following features:

a. Two missiles shall not be simultaneously armed.
b. A missile shall be armed only when the launcher is erect. Selection of load or stow mode when a RAD is armed shall result in the armed RAD driving to the safe state.
c. An armed missile shall be placed in the safe state if the missile select control is moved to OFF or to select another missile.
d. After missile firing, placing missile arm to the safe position shall enable circuitry to allow the MGS "wire-cut" signal to pass to the spent missile cannister.

3.6.15.4 Smoke grenade system. The vehicle shall accept a self-screening protective smoke system. When the smoke grenade launcher power control selector is ON, the indicator illuminated, and the firing switches activated, supply voltage at the launcher grenade terminals shall correspond to vehicle battery voltage.

3.6.15.5 Electromagnetic interference.

3.6.15.5.1 Electromagnetic interference characteristics. Unless otherwise specified (see 6.2), the vehicle shall be radio interference suppressed in accordance with classes A3 and C1 of MIL-STD-461.

3.6.15.5.2 Weapon station. The weapon station shall be compatible with the M113A1 systems and the TOW system components. The M901 weapon station shall conform to specified requirements for electrical/electro-mechanical equipment in MIL-STD-461, except for the limits specified in ATPD-l.

3.6.16 Manual back-up system. The weapon station shall provide a manual back-up mode. In the absence of primary electrical power to the turret, the manual back-up mode shall permit manual erection of the launcher, use of back-up emergency battery power sufficient to fire up to ten missiles in succession, and use of the manual hydraulic pump to maintain operating level of hydraulic system pressure.

3.6.16.1 Manual erect. The weapon station shall provide a manual stow-to-erect capability, using a hand crank which is inserted into the erection arm. The maximum torque required to erect the launcher shall not exceed 20 pound-feet. Engagement of the hand crank shall inhibit the operation of the erection drive motor.

3.6.16.2 Emergency power. The weapon station shall feature a back-up emergency power interface provision, which provides electrical power to the control panel assembly in the weapon station but not to the electrical hydraulic pump.

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3.6.16.3 Manual hydraulic pump. A manual hydraulic pump shall be provided, which shall enable an operator to achieve hydraulic system pressure of 1750 psi (minimum). The manual hydraulic pump shall enable the operator to raise the accumulator pressure from 1000 psi up to 1500 psi in 150 pump cycles, and the maximum required force shall not exceed 60 pounds.

3.6.17 Air transport and shipping configuration. The air transport configuration shall be achieved for the vehicle by removal of high stow supports and the acquisition sight and armor, and by manually cranking down the erection arm. In the transport configuration mode, the vehicle shall not exceed a gross vehicle height of 102.5 inches, when the chassis deck height is no greater than 72 inches.

3.6.18 Machine gun. The machine gun provision of the vehicle shall consist of a traversing rail and pintle mount for the 7.62mm M60 machine gun. The rail shall be mounted on the weapon station. The force applied at the pintle, required to move the machine gun mount along the traversing rail, shall not exceed 15 pounds after breakaway. A locking device shall be provided to secure the mount at any position along the traversing rail. The mount, locked in any position along the traversing rail, shall have a breakaway force not less than 45 pounds. In its travel mode the machine gun shall:

a. Not interfere with any position of the launcher or weapon station.
b. Permit continuous weapon station rotation without chassis interference.
c. Remain stowed under the conditions defined in the operational profile (see 3.2.3).

3.7 Noise level. With a sound level meter located 4 inches forward of the engine compartment bulkhead, 4 inches to the left of the engine compartment left bulkhead, and 12 inches below the inside surface of the top armor plate, and with the chassis traveling at 30 mph in 1-3 range on a hard-surfaced straight road with all hatches closed, the sound level shall not exceed 119 decibles (dba).

3.7.1 Noise hazard caution sign. The vehicle driver and passenger compartments shall be posted as noise hazardous areas in accordance with MIL-STD-1474.

3.8 Painting, marking and data plates.

3.8.1 Painting. Unless otherwise specified (see 6.2), exterior and interior of the vehicle, ancillary and stowage equipment, components assemblies and parts which require painting, shall be prepared and painted in accordance with MIL-STD-193 or TT-E-529. The exterior topcoat, interior of the vehicle ramp, and underside of the driver and gunner hatch shall be camouflage lusterless forest green enamel conforming to either MIL-E-52798 or MIL-E-52835. Interior of the vehicle, excluding those areas described herein, shall be painted light green number 24533 of FED-STD-595 in accordance with TT-E-529.

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3.8.2 Marking. Vehicle exterior markings shall be lusterless black color, number 37038 of FED-STD-595 except for safety markings, and shall conform to MIL-STD-642. Otherwise, the weapon station shall be marked in accordance with MIL-W-13855. Parts and components shall be marked in accordance with MIL-STD-130.

3.8.3 Safety markings. Unless otherwise specified (see 6.2), safety hazard plates and decals for the vehicle and component assemblies shall be posted in accordance with applicable drawings.

3.8.4 Name, shipping and service data plates. The plates shall meet the requirements of MIL-P-5l4, or applicable drawings.

3.9 Workmanship. Workmanship shall be of a quality to assure that the vehicle and its components are free of defects that compromise, limit or reduce capability in the performance of its intended use. In addition to general appearance, the defects listed in table IV shall be considered to be cause for failure. Electric wires and hydraulic lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. No workmanship condition shall exist which may create a safety hazard to operating or maintenance personnel.

3.10 TOW Dismount/remount. The TOW component mounting and storing provisions shall enable a four man crew to dismount the TOW components, needed for ground deployment, within 5 minutes for emplacement within 50 meters of the vehicle. The same crew shall mount TOW components and return the vehicle to a combat ready condition within 5 minutes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials and qualified products. The contractor's records shall be reviewed to determine conformance to 3.3, 3.3.1 and 3.3.2.

4.1.2 Parts and components. Parts, components and assemblies shall be inspected for conformance to requirements of drawings subordinate to 8736977 and applicable specifications and standards (see 3.4.1, 3.4.2, and 3.4.3). When applicable, inspection shall also be in accordance with Supplementary Quality Assurance Provisions (SQAPs) or Quality Assurance Requirements (QARs).

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4.2 Classification of inspections. Classification of inspection shall be as follows:

a. First article (preproduction) inspection (see 4.3).
b. Initial production vehicle inspection (see 4.4).
c. Quality conformance inspection (see 4.6).

4.3 First article (preproduction) inspection. One first article sample (see 3.1) shall be inspected, at a location approved by the Government, to determine conformance to requirements of this specification. Inspection shall consist of examination as specified in 4.6.2, and all tests specified herein.

4.3.1 Failure. Failure to pass any examination or test, or defects in excess of acceptable quality levels (AQLs) specified, shall be cause for refusal to grant first article approval.

4.4 Initial production vehicle inspection. On beginning production, (six) vehicles shall undergo, and shall pass, initial production inspection (see 3.2). One vehicle shall undergo first production vehicle inspection, and the second through sixth vehicle(s) will be subjected to the initial production test.

4.4.1 First production vehicle inspection.

4.4.1.1 In-process examination. During fabrication of the first production vehicle, an in-process examination will be conducted by representatives of TACOM Product Assurance Directorate to evaluate conformance of materials and workmanship to specified requirements. Examination will be made at the contractor's or subcontractor's facility prior to application of primer and paint. Processing and welding procedures, quality system, and inspection records will be evaluated during this examination.

4.4.1.2 Completed first production vehicle contractor inspection. The first completed production vehicle shall be road tested and inspected by the contractor, as specified in 4.6.5, to determine conformance to contract and specifications. After inspection, the contractor shall submit the vehicle (and all inspection records and certifications) to the responsible Government inspection element at contractor's plant for preliminary examination.

4.4.1.3 Preliminary examination. The responsible Government inspection element will conduct a preliminary examination, as specified in 4.6.2, of the first completed production vehicle.

4.4.1.4 Provisional inspection. Provisional inspection of the first completed production vehicle will be conducted jointly by representatives of TACOM Product Assurance Directorate and the responsible Government inspection element. Contractor shall provide any required assistance. Testing will be as specified in 4.6.5. Contractor shall make available his inspection plan, records, and certifications pertinent to the vehicle and components.

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4.4.1.5 Repair of defects. Defects found as a result of the foregoing inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to correct defects promptly shall be cause for suspension of acceptance of vehicles until corrective action has been approved by the Government.

4.4.1.6 Vehicle disposition. On completion of first production vehicle inspection, the vehicle shall remain at the manufacturing facility, as a production sample, and shall be the last vehicle shipped on contract. Vehicle may be released sooner at the discretion of the Government. The contractor shall service and maintain vehicle during this period.

4.4.1.7 Final approval and acceptance. Final approval and acceptance of the first production vehicle shall be withheld until the second through sixth vehicles(s) are accepted (see 4.4.2).

4.4.2 Second thru sixth initial production vehicles inspection. Five additional vehicles (see 4.4) from the first month's production, or the first 20 produced, will be subjected to the initial production test.

4.4.2.1 Initial production test. To determine conformance to Section 3 (inclusive), vehicles will be examined as specified in 4.6.2, and tested as specified in table V. Subsequently, the vehicles will be road tested as specified in 4.4.2.2. Tests will be performed by the Government at a site approved by the Government and will require no more than 120 days. Delays caused by vehicle breakdown due to poor quality of workmanship or material, or failure of the contractor to comply with specification or drawing requirements shall not be the basis for adjustment of the contract performance dates, delivery schedule or contract price. The contractor shall expeditiously furnish repair parts as required to support testing.

4.4.2.2 Test profile. Three test vehicles shall be subjected to 7 repetitions of the 750 mile cycle contained in table III for a total test mileage of 5250 miles. The weapon station shall be subjected to the number of simulated missions specified at the rate of one mission every 25 miles. At the mid-point of each 25 mile cycle, an accelerated silent watch operation shall be performed. Two test vehicles shall be subjected to the initial production tests outlined in table V and then to 3 repetitions of the 750 mile cycle contained in table III.

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TABLE III. 750 Mile test cycle.

Course	Mileage and speed 1/	Number of ITV missions
Hard-surfaced roads	150	6
Gravel and dirt roads	300	12
Level cross-country 2/	150	6
Hilly cross-country 2/	150	6

1/ Speeds shall be varying up to the maximum safe speed in all applicable gear ranges depending on course and conditions.

2/ Ten percent (10%) of all operation on cross-country courses shall be conducted with the launcher erected and a vehicle speed not to exceed 5 mph.

4.4.2.3 Reliability verification. To determine conformance to 3.2.1, a point estimate of the MTBF equal to greater than 180 hours must be demonstrated during the test. The point estimate will be computed by dividing the total weapon station hours (adjusted to 11.5 hours per mission), of the ITV undergoing durability testing, by the total cumulative chargeable combat mission failures.

4.4.2.4 Maintainability verification. To determine conformance to 3.2.2, the point estimate of the maintenance ratios for the ITV installed on the IPT durability vehicles shall not be greater than 0.05 for Organizational Maintenance, 0.04 for Direct Support Maintenance, and 0.09 for Combined Organizational and Direct Support. The maintenance ratio shall be calculated by dividing the active maintenance man hours (scheduled and unscheduled scored in accordance with the Improved TOW Vehicle RAM-D Failure Definition and Scoring Criteria experienced on the kits of these ITV's) for each level of maintenance by the total ITV kit hours, adjusted to 11.5 hours per mission.

4.4.2.5 Failure. Failure of the vehicle attributable to a deficiency in workmanship or materials during, or as a result of, the test shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the manufacturer that corrective action has been taken to eliminate the cause of the deficiency. Any deficiency found during or as a result of the test shall be prima facie evidence that all vehicles already accepted prior to completion of the IPT are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all vehicles shall be corrected by the contractor at no cost to the Government.

4.5 M901 System integration break-in. To determine conformance to 3.5, each vehicle shall be operated for a distance of fifty miles or more on smooth, hard-surfaced roads. A section of road shall be as shown in figure 1 and shall be traveled at least once every 5 miles. Except for equipment and component installation, and servicing performed by the contractor as a final processing of vehicles for acceptance, the vehicles shall be completely assembled. Vehicles

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shall be serviced with fuel, lubricants, and fluid as specified, but need not be combat loaded, except for those vehicles which have been designated control test vehicles by the responsible Government inspection representative (see 4.5.1). The automotive acceptance tests, speed, acceleration, and holding may be performed during this run. After the break-in run, the slope tests as specified in 4.7.20 shall be performed, the vehicle voltage generation acceptance test shall be performed, and the vehicle shall be examined for evidence of fuel, lubricant, hydraulic fluid leakage, damaged components or loss of original adjustment that may cause faulty operation subsequent to Government acceptance.

4.5.1 Control test vehicles. Control test vehicles shall be run with a simulated combat load which yields a total vehicle weight of 25,500 + 800 pounds.

4.6 Quality conformance inspection.

4.6.1 Inspection provisions.

4.6.1.1 Lot size. An inspection lot shall consist of all vehicles produced during an identifiable production period, and submitted at one time for acceptance inspection. Production period shall not exceed one month.

4.6.1.2 Sampling for inspection. For the purpose of visual, dimensional, and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling, the contractor shall 100 percent inspect the first 10 vehicles to establish a process average (see 6.5), to allow normal sampling in accordance with MIL-STD-105.

4.6.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to applicable drawings and this specification. Examinations shall be performed against the classification of defects and with the acceptable quality levels (AQLs) specified in table IV. Defects found during examinations shall be listed on the deficiency sheet for the vehicle (see 6.7). The following constitutes a part of the classification of defects (see 6.6):

a. Any weep constitutes a minor defect.
b. Any seep constitutes a minor defect.
c. Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
d. Any droplet that occurs at a static fit, metal to metal, or gasket combination shall constitute a major defect.

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TABLE IV. Classification of defects.

Category	Defect	Method of inspection
Major:	AQL 25 defects/100 units	
101	Engine; malfunction, leaks, mounting	Visual and functional
102	Transmission; malfunction, leaks.	Visual and functional
103	Transfer gear differential steer and final drive units: malfunction, leaks.	Visual and functional
104	Fuel system components: malfunction, leaks. 1/	Visual and functional
105	Cooling system components; malfunction, damage, leaks.	Visual and functional
106	Instrumentation, switches, warning, indicating and safety devices; malfunction, damage, calibrations.	Visual and functional
107	Electrical system components circuits, and communication system; malfunction, lighting sequence improper. 2/	Visual and functional
108	Suspension system, components; adjustments, damage, leaks, clearance improper, misalignment.	Visual and functional SIE 3/
109	Tracks; damage, pad separation.	Visual and SIE 3/
110	Doors, hatchs, covers and windows; fit improper malfunction, leaks, damage, non-conformance.	Visual and functional
111	Ramp and hydraulic components; improper door fit, malfunction, leaks, damage.	Visual and functional
112	Trim vane; malfunction	Visual and functional
113	Vision devices and receptacles; damage, malfunction.	Visual and functional
114	Bilge pumps; malfunction, damage	Visual and functional
115	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, and hardware; malfunction, locking devices, damage.	Visual and functional
116	Controls; malfunction, clearance improper.	Visual and functional
117	Track drive sprockets and drive shafts; damage, clearance improper.	Visual and SIE 3/
118	Adjustment mechanisms; malfunction, clearance improper.	Visual and functional
119	Hull; structural and welding defects	Visual and functional

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TABLE IV. Classification of defects. - Continued

Category	Defect	Method of inspection
120	Towing pintle, lifting eyes, hooks and brackets; malfunction, defects.	Visual and functional
121	Seats, crash pads, and straps; damaged, seat control malfunction.	Visual
122	Fire extinguisher system control; seals missing, components damage.	Visual
123	Wire, tubing or hoses; defective, damage.	Visual
124	Hydraulic control system, pump, motor reservoir, accumulator manifolds, tubing; malfunction,, damage, droplets.	Visual and functional
125	Weapon hatch; malfunction, torsional spring pop-up or open locking devices damage.	Visual and functional
126	Manual back-up hydraulic pump; damaged, malfunction, clearance improper.	Visual and functional
127	Weapon control panel; malfunction, damage, improper indicators.	Visual and functional
128	Optics system; malfunction, damage.	Visual and functional
129	ITV Launcher assembly, launch erection arm, cupola; malfunction, damage, clearance improper, structural and welding defects.	Visual and functional
130	Launch AZ and EL drive systems; malfunction, damage.	Visual and functional
131	ITV kit ancillary and stowage equipment; damage, clearance improper, structural and welding defects.	Visual and functional
132	Lubrication and lubricants; contamination.	Visual
Minor:	AQL 150 defects/100 units	
201	Engine; improper component assembly.	Visual and functional
202	Transmission; improper component assembly or installation.	Visual and functional
203	Transfer gear, differential steer and final drive units; improper component assembly or installation.	Visual and functional
204	Fuel system components; improper line clearance, assembly or installation. l/	Visual and functional
205	Cooling system components; improper assembly or installation.	Visual and functional
206	Instrumentation, switches, warning, indicating and safety devices; improper assembly or installation.	Visual and functional

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TABLE IV. Classification of defects. - Continued

Category	Defect	Method of inspection
207	Electrical system components, circuits, and communication system; improper assembly or installation. 2/	Visual and functional
208	Suspension system components and adjustments; improper installation or assembly, bolt or nut torque.	Visual and functional
209	Tracks; tension, installation or assembly improper, including bolt torque, damage.	Visual and SIE 3/
210	Doors, hatches, covers and windows; improper assembly or installation.	Visual and SIE 3/
211	Ramp and hydraulic components; improper assembly or installation, fluid level and type.	Visual and functional
212	Trim vane; improper assembly or installation.	Visual and functional
213	Vision devices and receptacles; improper assembly, installation or clearance.	Visual and functional
214	Bilge pumps; improper assembly, installation.	Visual and functional
215	Covers, grilles, ducts, vents, panels, stowage boxes, seats, fenders, and hardware: improper fits, adjustment, installation or defective welds, seals, adhesion, hardware or sheet metal.	Visual and functional
216	Controls; improper assembly, or installation.	Visual and SIE 3/
217	Track drive sprockets and drive shafts; improper assembly, installation or bolt torque.	Visual and SIE 3/
218	Adjustment mechanisms; improper assembly or installation.	Visual and functional
219	Towing pintle, lifting eyes, hooks and brackets; improper fit, installation assembly, or adhesion welding defects, damaged.	Visual and functional
220	Seats, crash pads, and straps; data card incomplete, improper assembly or installation, fit or adhesion, welding defects.	Visual and functional
221	Fire extinguisher system control; data card incomplete, improper assembly, installation, coding or protection.	Visual
222	Wiring, tubing or hoses; defective, missing, improper installation, coding or protection.	Visual
223	Paint; application or color improper.	Visual
224	Tube fittings; defective, missing, improperly installed.	Visual
225	Decals, marking data and instruction plates; missing, improper location or size, defective.	Visual

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TABLE IV. Classification of defects. - Continued

Category	Defect	Method of inspection
226	Lubrication and lubricants; improper application, types and levels.	Visual
227	Protective and anti-skid coatings; application, materials and coverage improper.	Visual
228	Record forms, and publications; missing, improperly processed or secured.	Visual
229	Manual backup, hydraulic pump; improper installation.	Visual and functional
230	Hydraulic control system, pump, motor reservoir, accumulator manifolds, tubing improper assembly, installation, fluid level, precharge pressure.	Visual and functional
231	Weapon hatch; improper fits, adjustment, installation, defective welds, seals, pressure switch installation.	Visual and functional
232	Weapon control panel; improper assembly or installation.	Visual and functional
233	Optics system; improper assembly, installation, dessicant.	Visual and functional
234	ITV Launcher assembly, launch erection arm, cupola; improper assembly or installation.	Visual and functional
235	Launcher AZ and EL drive systems; improper assembly or installation.	Visual and functional
236	ITV kit ancillary and stowage equipment; improper assembly, fit, installation as per drawing requirements.	Visual and functional
237	Fasteners; improper assembly, torque.	Visual and functional

1/ Fuel leaks constitute a special defect and will result in total inspection of the lot for this defect.
2/ Filament failures are not to be classified as major or minor defects.
3/ SIE means Standard Inspection Equipment.

4.6.2.1 Unclassified defects. Defects considered to be departures from good workmanship but having no bearing on function, safety, interchangeability, or life shall be noted in writing. Such defects recurring in five consecutive lots, or ten lots or more within a thirty day period, will be added to the minor classification of defects with no increase in AQL. A workmanship defect may be dropped from the minor classification of characteristics when five consecutive lots are found free of the defect.

4.6.2.2 Examination failure. If a vehicle fails to pass any examination specified herein, the Government will stop acceptance until evidence has been provided by the contractor that corrective action has been taken.

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4.6.3 Classification of tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.6.4).
b. Control tests (see 4.6.5).
c. Comparison tests (see 4.6.6).

4.6.4 Acceptance tests. To determine conformance to Section 3 (inclusive), the contractor shall subject each vehicle to the acceptance tests specified in table V. After the tests, the vehicle shall be examined for evidence of fuel or lubricant leakage, or other defects. All defects found during tests shall be listed on the deficiency sheet for the vehicle (see 6.7).

4.6.4.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government will withhold acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective action has been taken.

4.6.5 Control tests.

4.6.5.1 Frequency. One of the first 10 vehicles produced, and thereafter one per month, sha11 be selected for control testing after break-in run.

4.6.5.2 Fifty mile test. Vehicle shall be examined as specified in 4.6.2, loaded with a simulated combat load to a total weight of 25,500 + 800 pounds, and operated for a distance of 50 miles over relatively smooth, hard-surfaced roads. During and subsequent to the test, as appropriate, vehicle shall meet the requirements specified for control tests in table V.

4.6.5.3 Failure. If the vehicle fails to pass any control test specified herein, the Government will withhold acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective action has been taken.

4.6.6 Comparison tests. The Government may select vehicles at any time during the contract production period and subject them to the examinations specified in 4.6.2 and tests specified in table V to compare existing vehicle quality with previous standards and to reveal defects of a workmanship or materials nature that may reduce the effective operation of the vehicle in the field. These tests will be conducted at Government laboratories or proving grounds designated by the contracting officer. Vehicles selected for comparison tests will be combat loaded with all BII stowed in their respective spaces and shall not include vehicles previously tested.

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4.6.6.1 Test failure. Failure of any M901 vehicle tested to conform to any of the requirements specified in the contract, or any major defects of a workmanship or materials nature occuring during, or as a result of, the test cycle, may be considered cause for refusal to continue acceptance of vehicles by the Government until objective action has been taken to eliminate the cause of failure. Any defect found during or as a result of the test shall be prima facie evidence that vehicles accepted subsequent to the accepted initial production test vehicle or previous acceptable comparison test vehicles are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly defective. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

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TABLE V Classification and location of inspections and tests.

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Reliability	3.2.1	4.4.2.3			
Maintainability	3.2.2	4.4.2.4			
Seals, weather	3.4.4.1	4.7.1	4.7.1	4.7.1	
Seals, bearings	3.4.4.2	4.7.2	4.7.2		4.7.2
Hatches and vision devices	3.4.5	4.7.1	4.7.1		
Hull periscopes and vision block	3.4.5	4.7.40	4.7.40		
Electrical circuits	3.4.6.1	4.7.3.1	4.7.3.1		
Generating system voltage	3.4.6.2	4.7.3.2	4.7.3.2		
Lights	3.4.6.3	4.7.3.3	4.7.3.3		
Adjustment mechanisms	3.4.7	4.7.4	4.7.4		
Welding repairs and burning	3.4.8	4.7.5	4.7.5		
Controls	3.4.9	4.7.6	4.7.6		
Nightsight control	3.4.9	4.7.39		4.7.39	
Hydraulic system	3.4.10	4.7.7	4.7.7		
Weight	3.4.11	4.7.8			4.7.8
Stowed equipment	3.4.12	4.7.9		4.7.9	4.7.9
Climatic conditions	3.6.1.1	4.7.10			4.7.10
Shock and vibration	3.6.1.2	4.7.10			4.7.10
Bilge pump operation	3.6.2	4.7.11		47.11	
Speed	3.6.3	4.7.12	4.7.12		4.7.12
Acceleration	3.6.4	4.7.13	4.7.13		4.7.13

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TABLE V Classification and location of inspections and tests. - Continued

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Stopping and vehicle control	3.6.5.1	4.7.14	4.7.14		4.7.14
Holding	3.6.5.2	4.7.15	4.7.15	4.7.15	
Shallow water fording and floatation	3.6.6.1	4.7.16	4.7.16		
Amphibious operation	3.6.6.2	4.7.17		4.7.17	4.7.17
Trench crossing	3.6.7	4.7.18			4.7.18
Vertical obstacle	3.6.8	4.7.19			4.7.19
Climbing	3.6.9	4.7.20	4.7.20		
Slope	3.6.10	4.7.20	4.7.20		4.7.20
Ramp system	3.6.11	4.7.21	4.7.21		
Power	3.6.12.1.1	4.7.22.1	4.7.22.1		
Launcher erection	3.6.12.2	4.7.22	4.7.22		
Load to erect	3.6.12.2	4.7.22.1	4.7.22.1		
Erection locks during manual procedures	3.6.12.2	4.7.22.2	4.7.22.2		
Manual erection	3.6.12.2	4.7.22.3		4.7.22.3	
Erection locks during manual procedures	3.6.12.2.1	4.7.22.2			
Manual erection	3.6.12.2.1	4.7.22.3		4.7.22.3	
Erection motor speed and time	3.6.12.2.2	4.7.22	4.7.22		
Erect to stow mode	3.6.12.3	4.7.23	4.7.23		
Launcher stow mode	3.6.12.3	4.7.23.1	4.7.23.1		
Load mode	3.6.12.4	4.7.24	4.7.24		
Azimuth/elevation positional capability	3.6.12.5	4.7.25	4.7.25		
Azimuth and elevation drives	3.6.12.6	4.7.25	4.7.25		

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TABLE V Classification and location of inspections and tests. - Continued

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Azimuth drive	3.6.12.6.1	4.7.25.1	4.7.25.1		
Slew speed range	3.6.12.6.1.1	4.7.25.1	4.7.25.1		
Track speed range	3.6.12.6.1.2	4.7.25.1	4.7.25.1		
Elevation rate	3.6.12.6.2	4.7.25.2	4.7.25.2		
Target tracking	3.6.12.6.3	4.7.26	4.7.26		
Hydraulic power supply	3.6.12.7.1	4.7.27	4.7.27		
Hydraulic system fluid cleanliness	3.6.12.7.2	4.7.27.1		4.7.27.1	
Pressurization, hydraulic system	3.6.12.7.3	4.7.7	4.7.7		
Azimuth/elevation operation	3.6.12.7.3	4.7.28	4.7.28		
Safety interrupt, erect launcher	3.6.12.8.1				
Weapon hatch open interrupt override	3.6.12.8.1	4.7.29.1	4.7.29.1		
Driver's hatch open interrupt	3.6.12.8.1	4.7.29.2	4.7.29.2		
Cargo hatch open interrupt	3.6.12.8.1	4.7.29.3	4.7.29.3		
Azimuth fire interrupt region	3.6.12.8.1	4.7.29.4	4.7.29.4		
Elevation fire interrupt	3.6.12.8.1	4.7.29.5	4.7.29.5		
Weapon hatch interlock	3.6.12.8.1	4.7.30	4.7.30		
Interrupt circuitry	3.6.12.8.1	4.7.31	4.7.31		
Erect to load circuitry	3.6.12.8.1	4.7.32	4.7.32		
Safety interrupt - Stow positioned launcher	3.6.12.8.2				
Weapon hatch interlock	3.6.12.8.2	4.7.30	4.7.30		
System interlock	3.6.12.8.2	4.7.33	4.7.33		
Safety interrupt - Load positioned launcher	3.6.12.8.3				

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TABLE V Classification and location of inspections and tests. - Continued

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Load to erect	3.6.12.8.3	4.7.22.1	4.7.22.1		
Load mode	3.6.12.8.3	4.7.24	4.7.24		
Control panel assembly function	3.6.12.9	4.7.34	4.7.34		
Control panel controls and indicators	3.6.12.9.1	4.7.34	4.7.34		
Missile umbilical continuity	3.6.12.9.2	4.7.37	4.7.37		
Tracking drive deadband	3.6.12.9.2.1	4.7.38		4.7.38	
Azimuth control	3.6.12.9.2.2	4.7.6	4.7.6		
Elevation control	3.6.12.9.2.3	4.7.6	4.7.6		
Hand controller indicator	3.6.12.9.2.4				
Nightsight remote controls	3.6.12.9.3	4.7.39		4.7.39	
Image transfer assembly (ITA)	3.6.13.1	4.7.36	4.7.36		

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TABLE V Classification and location of inspections and tests. - Continued

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Sight selection (ITA)	3.6.13.1.1	4.7.36	4.7.36		
Image transfer assembly (ITA) image quality	3.6.13.1.2	4.7.41	4.7.41		
Image quality movement (ITA)	3.6.13.1.3	4.7.41	4.7.41		
Squad leader periscope	3.6.13.2	4.7.42	4.7.42		
Daysight/tracker (MGS) interface	3.6.14.1	4.7.43	4.7.43		
Nightsight	3.6.14.2	4.7.39		4.7.39	
Launcher tubes boresight	3.6.14.3	4.7.44	4.7.44		
Launcher tubes boresight retention	3.6.14.3	4.7.45			4.7.45
Weapon station intercom noise	3.6.15.1.1	4.7.46	4.7.46		
Slipring noise	3.6.15.2	4.7.46	4.7.46		
Missile select/arm circuitry	3.6.15.3.1	4.7.35	4.7.35		
Missile select/arm/safe circuitry	3.6.15.3.2	4.7.35	4.7.35		
Missile umbilical continuity	3.6.15.3.2	4.7.37	4.7.37		
System interlock	3.6.15.3.3	4.7.33	4.7.33		
Missile select/arm circuitry (safety provisions)	3.6.15.3.3	4.7.35	4.7.35		
Missile umbilical continuity	3.6.15.3.3	4.7.37	4.7.37		

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TABLE V Classification and location of inspections and tests. - Continued

Title	Requirements	First article 1/ initial production tests	Acceptance tests 100 percent 2/	Control tests 2/	Comparison tests 1/
Smoke grenade system	3.6.15.4	4.7.47	4.7.47		
Electromagnetic interference (EMI)	3.6.15.5.1	4.7.48			4.7.48
Weapon station (EMI)	3.6.15.5.2	4.7.48			4.7.48
Back-up missile firing circuitry	3.6.16	4.7.49	4.7.49		
Manual erect	3.6.16.1	4.7.22.3		4.7.22.3	
Emergency power	3.6.16.2	4.7.49	4.7.49		
Azimuth/elevation operation	3.6.16.3	4.7.28	4.7.28		
Manual hydraulic pump	3.6.16.3	4.7.50		4.7.50	
Air transport	3.6.17	4.7.51			4.7.51
Machine gun mount and mechanical function	3.6.18	4.7.52		4.7.52	
Noise level	3.7	4.7.53			4.7.53
Noise hazard sign	3.7.1	4.7.53			4.7.53
TOW dismount/remount	3.10	4.7.54			4.7.54

1/ Government proving ground.
2/ Manufacturer/contractor facility.

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4.7 Conformance verification.

4.7.1 Weather seal leak test. To determine conformance to 3.4.4.1 and 3.4.5, all hatches, vision blocks and other installed vehicle components, such as the launcher, erection arm, periscopes, smoke grenade launchers, and ramp access doors shall be operated and examined for functional requirements. With all hatches closed, and vision blocks and other components properly installed, a spray of water shall be directed on the hatches, vision blocks and other components. Spray shall be applied from an open hose, with a line pressure of not less than 35 psi, at a rate of 10 gallons per minute for a period of 3 minutes. Leakage shall not exceed an average of 10 drops per minute collectively.

4.7.2 Bearing seals examination. To determine conformance to 3.4.4.2, prior to and after fording and amphibious operations, the vehicle shall be examined for lubricant leakage and water contamination.

4.7.3 Electrical system tests.

4.7.3.1 Circuits test. To determine conformance to 3.4.6.1, electrical circuits supplying power to vehicle equipment shall be tested to assure continuity and integrity. Circuits tested for continuity prior to vehicle installation and weapon station power circuitry and circuit functions shall be verified by evidence of functional performance during operational tests. The vehicle radio shall be operated to verify transmission and reception, and it shall be verified by evidence of operational performance that the ITV electrical system is compatible with the vehicle power supply, which conforms to requirements of MIL-STD-1275.

4.7.3.2 Generating system voltage (alternator) test. To determine conformance to 3.4.6.2, with the alternator running at 4000 rpm and producing 50 $\pm$ 5 amperes, a final voltage test shall be made with a voltmeter at the vehicle- battery input connector cable.

4.7.3.3 Light test. To determine conformance to 3.4.6.3, all vehicle lighting, including running lights, driver's console lights, turret level and driver level illumination, azimuth scale illumination, and weapon station indicators and brightness controls, shall be operated during and after road testing through all stages of operation.

4.7.4 Adjustment mechanisms test. To determine conformance to 3.4.7, all adjustment mechanisms shall be observed during operation, examined for proper adjustment and adjusted, if required.

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4.7.5 Welding repairs and burning test. To determine conformance to 3.4.8, all welding repairs and burning operations shall be examined to determine that slag has been removed and approved procedures have been followed.

4.7.6 Controls test. To determine conformance to 3.4.9, 3.6.12.9.2.2 and 3.6.12.9.2.3, all electrical, mechanical and hydraulic controls shall be operated and functional performance demonstrated. The hand controller shall be moved in azimuth/elevation to verify no indication of binding through maximum deflections in azimuth/elevation and smoothness, and continuous mechanical operation of the nightsight controls shall be demonstrated.

4.7.7 Hydraulic system test. To determine conformance to 3.4.10, and 3.6.12.7.3 the hydraulic systems shall be operated, examined for leakage in excess of a "seep" condition (see 6.6), the hydraulic reservoir examined for correct fluid level, and it shall be verified that the accumulator precharge pressure is not less than 900 psi nor greater than 1100 psi. It shall be verified that internal leakage and resultant system pressure loss does not exceed specified limits by verifying that the time interval between hydraulic pump motor activation when the M90l weapon station is powered up but not active, is not less than 7 minutes. System shall have been temperature stabilized prior to examining for internal leakage.

4.7.8 Weight test. To determine conformance to 3.4.11, the combat loaded vehicle shall be weighed.

4.7.9 Stowed equipment test. To determine conformance to 3.4.12, all manufacturer and depot installed BII, including the hard equipment items listed in 3.4.12.1, shall be stored (combat loaded) on the selected vehicle. All interior and exterior items shall fit in spaces provided, and shall not interfere with operation of the vehicle and components in any way. The equipment shall be removed after the test. The BII used shall be of the latest production available. It shall be permissible to unpackage and repackage equipment used for test.

4.7.10 Environmental tests. To determine conformance to 3.6.1.1 and 3.6.1.2, the vehicles shall be operated in specified environments and observed for performance requirements.

4.7.11 Bilge pump operational test. To determine conformance to 3.6.2, with sufficient water in the vehicle hull, the bilge pumps shall be operated and observed for functional requirements.

4.7.12 Speed test. To determine conformance to 3.6.3, the vehicle shall be driven at speeds up to the specified maximum and observed for performance requirements.

4.7.13 Acceleration test. To determine conformance to 3.6.4, the vehicle shall be operated as specified to determine acceleration time from 0 to 20 mph.

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4.7.14 Stopping and vehicle control test. To determine conformance to 3.6.5.1, the vehicle shall be driven and stopped, as specified, a minimum of three times. Stopping distance of the three stops shall be averaged to determine the final result.

4.7.15 Holding test. To determine conformance to 3.6.5.2, the vehicle shall be driven up the specified grade, brakes applied and vehicle observed for movement, then with brakes locked for parking, the vehicle shall be observed for movement.

4.7.16 Shallow water fording and floatation test. To determine conformance to 3.6.6.2, the vehicle shall be driven into water of specified depth without special equipment. Water accumulation shall not exceed specified amount after 15 minutes of test.

4.7.17 Amphibious operation test. To determine conformance to 3.6.6.2, the vehicle shall be operated amphibiously and performance observed. The amount of water in the hull and leakage of lubricants from the bearings shall be measured.

4.7.18 Trench crossing test. To determine conformance to 3.6.7, the vehicle shall be operated over trenches of specified depth and width.

4.7.19 Vertical obstacle test. To determine conformance to 3.6.8, the vehicle shall be operated over vertical obstacles of specified height.

4.7.20 Climbing, grades, and slopes test. To determine conformance to 3.6.9 and 3.6.10, after completion of the M901 system integration break-in (see 4.5), the vehicle shall be tested for its specified operational capabilities on a 60 percent forward and rear grade and on a 30 percent left and right lateral slope. Azimuth and elevation positions shall be evaluated by proof of conformance to table VI.

4.7.21 Ramp system test. To determine conformance to 3.6.11, the ramp hydraulic system shall be operated and examined for evidence of leaks. The ramp shall be operated to fully opened and closed positions and observed for functional requirements.

4.7.22 Launcher erection test. To determine conformance to 3.6.12.2, and 3.6.12.2.2, the launcher shall be erected and it shall be verified that erection time is 8 seconds or less. During hydraulic pump operation, peak current shall be 900 amperes or less and steady state current shall not exceed 200 amperes. It shall be visually verified that the launcher decelerates prior to achieving the erect position, that erection locks are in position, that launcher erect and ready indicators come on, and that AZ and EL stow indicators are in conformance with table VI.

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4.7.22.1 Load to erect test. To determine conformance to 3.6.12.1.1, 3.6.12.2, and 3.6.12.8.3, the load to erect circuitry shall be verified by opening the cargo hatch and observing successful operation of the erection drive to erect the launcher. It shall be visually verified that the launcher decelerates prior to achieving the erect position, that erection locks deploy and appropriate CPA indicators come on in accordance with table VI.

4.7.22.2 Erection locks during manual procedures. To determine conformance to 3.6.12.2, and 3.6.12.2.1, it shall be verified that erection lock cylinders remain engaged when hydraulic pressure is bled down to the precharge level specified in 3.6.12.7.3 and all ITV weapon station power switches are set to “OFF”. The manual hydraulic pump shall be used to raise system indicated pressure to about 1500 psi, and it shall be verified that erection locks retract and that locks then engage when emergency power is turned to “ON”.

4.7.22.3 Manual erection test. To determine conformance to 3.6.12.2, 3.6.12.2.1, and 3.6.16.1, the launcher shall be put in the stow position and manually cranked up to erect position. The maximum torque required shall be less than 20 pound-feet. It shall be verified that after accumulator pressure has been manually pumped to 1200 psi or greater, emergency weapon station power “ON”, and launcher position select in “ERECT”, that erection locks engage when the launcher erect position is achieved.

4.7.23 Erect to stow test. To determine conformance with 3.6.12.3, stow circuitry, including enable and interrupt circuitry and proper CPA indicators shall be verified. Launcher position indicators shall illuminate in accordance with table VI.

4.7.23.1 Launcher stow test. To determine conformance to 3.6.12.3, it shall be verified that all stow indicators are in accordance with table VI, and that the launcher is resting on its STOW supports.

4.7.24 Load mode tests. To determine conformance to 3.6.12.4 and 3.6.12.8.3, it shall be verified that for CPA AZ and EL indicators – ON and WPM hatch indicator – OFF, the launcher can be driven to the load position. Using an inclinometer, the erection arm angle difference from erect to load shall be measured. It shall be verified that the "ready to load" indicator comes ON with the launcher in the load position. Verify that neither left nor right launch tube RADS can be armed with the launcher in load position.

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TABLE VI. Launcher position controls indications.

Launcher position	Illuminated indication								
	Turret power	Launcher erect	Launcher ready	AZ Load	Elevation load	Load position	AZ Stow	Elevation stow	Stow
ERECT	X	X	X						
ERECT - Launcher at stow enable AZ and EL	X	X	X				X	X	
ERECT - Stow selected, ready to drive into stow	X	X					X	X	
STOW	X						X	X	X
ERECT - Launcher at load enable AZ and EL	X	X	X	X	X				
ERECT - Load selected ready to drive into load	X	X		X	X				
LOAD	X			X	X	X			

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4.7.25 Azimuth/elevation positional capability test. To determine conformance to 3.6.12.5 and 3.6.12.6, the hand controller shall be used to rotate the M901 weapon station 360 degrees right and left, and to demonstrate maximum elevation and depression of the launcher.

4.7.25.1 Azimuth drive test. To determine conformance to 3.6.12.6.1, 3.6.12.6.1.1 and 3.6.12.6.1.2, the hand controller action controls shall be used to demonstrate azimuth drive control in the track speed range and the slew speed range. Maximum slew rate shall be verified by measuring time to travel 180 degrees, when hand controller is deflected to the maximum displacement in traverse. This shall be done in both directions and the hydraulic system accumulator must be fully charged initially, without the pump motor running.

4.7.25.2 Elevation rate test. To determine conformance to 3.6.12.6.2, the hand controller shall be used to demonstrate the range control of launcher elevation rate. Maximum launcher elevation and depression rates shall be verified by measuring time to travel a predetermined increment of launcher elevation angle, with maximum hand controller displacement.

4.7.26 Target tracking test. To determine conformance to 3.6.12.6.3, four tracking trials shall be performed by a qualified operator for 5 MRAD/second crossing target left to right and right to left for a target moving, both up and down, at 1 MRAD/second in elevation. The operator shall demonstrate 85 percent time-on-target during each trial. Crossing target trials shall be a minimum of 8 seconds duration, and vertical moving targets shall be a minimum of 4 seconds duration. At least 1 second shall be spent tracking the target crossing at .1MRAD/second or less during the vertical motion trials. Should the tracking performance be suspected of not conforming to specification, the trials may be repeated up to 3 times. For cases in which the tester is unable to judge conformance, provision shall be made to determine accurately the actual time-on-target during the tracking trials.

4.7.27 Hydraulic power circuit test. To determine conformance to 3.6.12.7.1, it shall be verified, when the accumulator pressure is less than 1400 psi and the CPA power is turned on, that the hydraulic pressurization pump comes on, and the pump turns off when the accumulator pressure reaches 1800 $\pm$ 100 psi.

4.7.27.1 Hydraulic system cleanliness test. To determine conformance to 3.6.12.7.2, a sample of hydraulic fluid from the weapon station hydraulic system shall be tested for conformance to the specified level of cleanliness after completion of the M901 system integration break-in run and prior to completion of the acceptance tests.

4.7.28 Azimuth/elevation operation test. To determine conformance to 3.6.12.7.3 and 3.6.16.3, the weapon station hydraulic system shall be manually pumped to meet the requirements, and it shall be verified that the launcher can be slewed in azimuth and elevation using only the manual back-up hydraulic pump and emergency power.

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4.7.29 Interrupt circuitry tests.

4.7.29.1 Weapon hatch open interrupt override test. To determine conformance to 3.6.12.8.1, the fire interrupt circuitry shall be verified by opening the weapons hatch and observing that fire interrupt and WPM hatch indicators are on, and the disabling of the trigger signal and AZ and EL action. The fire interrupt alarm shall be demonstrated, and the override function shall enable a trigger signal and permit action in AZ and EL. It shall be verified that closing the weapon hatch turns off the fire interrupt alarm and permits normal system operation.

4.7.29.2 Driver's hatch open interrupt test. To determine conformance to 3.6.12.8.1, the fire interrupt circuitry shall be verified by opening the driver's hatch, and observing that the fire and driver's batch interrupt indicators are – ON. Verify that the override function enables a trigger signal and permits action in AZ and EL.

4.7.29.3 Cargo hatch open interrupt test. To determine conformance to 3.6.12.8.1, the fire interrupt circuitry shall be verified by opening the cargo hatch (not fully-open position), and observing that the CPA fire interrupt and cargo indicators are ON. Verify that the override function enables a trigger signal and permits action in AZ and EL.

4.7.29.4 Azimuth fire interrupt region test. To determine conformance to 3.6.12.8.1, the azimuth angular regions of fire interrupt for launcher elevation, fully depressed, shall be tested. Azimuth regions are defined for the M901 vehicle in table II. Verify that with override on, missile firing shall be permitted in the fire interrupt region by monitoring the missile fire circuit breaker in the missile umbilical continuity test device.

4.7.29.5 Elevation fire interrupt test. To determine conformance to 3.6.12.8.1, the weapon station shall be driven through 360 degrees in azimuth and the elevation fire interrupt angles shall be tested in each azimuth region specified in table II by measuring launcher depression angle when the fire interrupt light comes on. Depression angles shall conform to table II.

4.7.30 Weapon hatch interlock test. To determine conformance to 3.6.12.8.1 and 3.6.12.8.2, the stow to erect circuitry shall be verified by opening the weapon station hatch sufficiently to activate the proximity switch and observing for the appropriate CPA indications. Verify that the launcher may be erected while the weapon station hatch is open.

4.7.31 Interrupt circuitry test. To determine conformance to 3.6.12.8.1, it shall be verified that the fire interrupt and WPM hatch indicators come on, the alarm is sounded, and action is inhibited in azimuth and elevation drive when the weapon station hatch has been opened.

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4.7.32 Erect to load circuitry tests. To determine conformance to 3.6.12.8.1, the launcher shall be maneuvered to the "LOAD" enable position of zero azimuth and maximum elevation. It shall be verified that the CPA indicators display as indicated in table VI. The angles shall then be measured. It shall be visually verified that when load mode is selected, the erection locks disengage and that, for a weapon station hatch open condition, the erection arm drive (to load) is inhibited.

4.7.33 System interlock test. To determine conformance to 3.6.12.8.2 and 3.6.15.3.3, it shall be verified that, with the launcher in the stowed mode, neither missile tube can be selected nor armed.

4.7.34 Control panel assembly function. To determine conformance to 3.6.12.9 and 3.6.12.9.1, 24 Vdc test power shall be supplied to the Control Panel Assembly (CPA) and all CPA indicator lights. Power – on circuitry, brightness controls, circuit breaker functions, and stow mode indications shall be verified.

4.7.35 Missile select/arm circuitry tests. To determine conformance to 3.6.15.3.1, 3.6.15.3.2 and 3.6.15.3.3, the missile select/arm circuitry for left and right launch tubes in the erect mode shall be verified. It shall be verified, on the Control Panel Assembly (CPA) indicators and direct inspection of the remote arming devices (RAD), that neither missile may be armed unless selected on the CPA.

4.7.36 ITA controls test. To determine conformance to 3.6.13.1 and 3.6.13.1.1, transfer assembly (ITA) sight controls shall be verified by selecting the 3X, 13X and night options, and verifying optics transfers within specified time.

4.7.37 Missile umbilical continuity test. To determine conformance to 3.6.12.9.2, 3.6.15.3.2 and 3.6.15.3.3, electrical continuity between the MGS and the left and right launch tube umbilicals shall be verified. Using an item of ITV peculiar support equipment, the missile umbilical test device, the missile error signals and the polarity at the remote arming device shall be verified. It shall be verified that the pre-fire, fire and wire cut signals pass from the MGS to the test round. Hand controller trigger indicator operation shall be verified.

4.7.38 Tracking drive deadband test. To determine conformance to 3.6.12.9.2.1, the weapon station control handle shall be deflected to 4 degrees in azimuth, the action switches shall be depressed and the weapon station azimuth angle indicator shall reveal weapon station motion. Test shall be repeated in the opposite direction in azimuth, and similarly for elevation and

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depression, and the launcher shall move for both forward and rearward 4 degrees control handle deflection when the action switches are depressed.

4.7.39 Nightsight tests. To determine conformance to 3.4.9, 3.6.12.9.3 and 3.6.14.2, the nightsight control linkages shall be connected and the weapon station nightsight controls exercised through their full range to verify smoothness of operation without binding. Linkages shall then be disconnected. Test shall be performed a total of ten times. Effective operation with the nightsight and controls shall be demonstrated by acquisition of a simulated night time target.

4.7.40 Hull periscopes and vision block test. To determine conformance 3.4.5, all vehicle hull periscopes and vision blocks shall be tested for fit and functional requirements.

4.7.41 ITA image quality test. To determine conformance to 3.6.13.1.2 and 3.6.13.1.3, the ITA shall be visually examined by a test gunner before and after tracking smoothness test to verify that the ITA exhibits no random motion of the image during launcher tracking, which impairs performance, and that there is no evidence of internal moisture and fogging.

4.7.42 Squad leader's periscope test. To determine conformance to 3.6.13.2, the squad leader's periscope shall be tested to verify that the specified articulations of the line-of-sight are maintained.

4.7.43 MGS interface tests. To determine conformance to 3.6.14.1, the standard set of missile guidance set (MGS) self-test 1 through 7 shall be performed as specified in MIS-14061. It shall be established that launcher elevation movements up and down result in MGS EL meter deflections right and left, and that launcher azimuth motion CW and CCW results in MGS AZ meter deflection right and left, respectively.

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4.7.44 Launcher tubes boresight test. To determine conformance to 3.6.14.3, the boresight elevation and azimuth angles of the launch tubes relative to the optical line of sight shall be tested using ITV peculiar support equipment, boresight target board, boresight test aperture and the boresight adapter tool. The test procedures listed in 4.7.45.1, shall be followed to assure that these angles are within specified tolerances.

4.7.45 Launcher tubes boresight retention test. To determine conformance to 3.6.14.3, the boresight angles of the launch tubes shall be tested using ITV peculiar support equipment, boresight target board, test aperture and adapter tool. To assure that boresight angles are within specified tolerances and procedure, the test procedures specified in 4.7.45.1 shall be followed for launcher elevation angles near maximum depression and elevation, and mid-range.

4.7.45.1 Launch tube boresight test procedure.

a. Apply power to the turret and erect launcher.
b. Install and test daysight/tracker, as applicable.
c. Insert drawing 12277611 boresight adapter and remote arming device alignment tool into right or left launcher in the same way a missile is loaded. Position RAD in the "ARMED" state. Insert shank of telescope boresight into socket of adapter with light finger pressure.
d. Open cargo hatch cover.
e. Set SELF TEST switch on Missile Guidance Set (MGS) to position 7 and depress OPERATE switch. Adjust azimuth and elevation knobs on daysight/tracker, and for mid-scale reading on MGS ELEVATION and AZIMUTH self test meters.
f. Close cargo hatch cover.
g. Set up drawing 12277600 target board at a distance of 100 feet 6 inches from the turret elevation axis.
h. Fit drawing 12277608 pinhole cover over daysight/tracker objective lens.
i. Set turret power switch on control panel to ON.
j. Set 3X, l3X, WFOV switch on turret hand controller to 13X.
k. Sight through turret eyepiece and operate turret to position sight on target.
l. Rotate the target board and center the vertical strip about the daysight vertical reticle. Secure the target board in this position.
m. Operate the turret to position daysight/tracker reticle on target board reticle.
n. Sight through the boresight telescope and verify that the crosshair reticle of the boresight is within the launch tube boresight launcher field check window on the target board.
o. Boresight the other launch tube in the same way.

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4.7.46 Communications. To determine conformance to 3.6.15.1.1 and 3.6.15.2, it shall be verified that each intercom station has talk/listen capability over both radio and intercom channels, in accordance with system performance requirements, over 360 degrees of weapon station azimuth rotation. Communications system noise level shall be monitored by ear during intercom use while the weapon station hydraulic system pump is cycled. If it is determined that excessive noise may be present, the intercom noise level shall be measured and shall be less than 350 millivolts AC-RMS when the hydraulic pump motor is cycling during 360 degrees of azimuth rotation.

4.7.47 Grenade launcher test. To determine conformance to 3.6.15.4, the grenade launcher power selector shall be placed "ON" and the indicator illumination shall be verified. The voltage between grenade terminals on the grenade launcher shall be measured as the firing switches are actuated.

4.7.48 EMI test. To determine conformance to 3.6.15.5.1 and 3.6.15.5.2, the vehicle shall be tested in accordance with MIL-STD-462.

4.7.49 Back-up missile firing circuitry test. To determine conformance to 3.6.16 and 3.6.16.2, the back-up power missile select/arm circuitry shall be verified by loading and using the ITV peculiar support equipment item, the missile umbilical continuity test round. Missile selections, arm, and firing functions shall be verified both visually at the round, with test indicator which shows RAD deployment, and in the gunner station, at least ten times.

4.7.50 Manual hydraulic pump test. To determine conformance to 3.6.16.3, the manual hydraulic pump shall be operated to raise accumulator pressure from 1000 psi to 1500 psi, in no greater than 150 pump cycles, and the maximum force required shall be demonstrated to be no greater than 60 pounds.

4.7.51 Air transport test. To determine conformance to 3.6.17, the specified components shall be removed from the launcher and the erect launcher shall be manually cranked down into a low stow transport position. Resultant height of the vehicle shall be measured. Prior to cranking, the launcher elevation angle shall first be depressed 5 degrees below horizontal.

4.7.52 Machine gun mount test. To determine conformance to 3.6.18, the force applied parallel to the traversing rail, at the machine gun mount pintle, which sustains motion after breakaway, shall be measured to be 15 pounds or less. It shall be verified that the locking device secures the mount at any position along the traversing rail, and that the mount will not move with a force of up to 45 pounds applied as specified.

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4.7.53 Noise test. To determine conformance to 3.7 and 3.7.1, it shall be verified, during tracking/slewing operations, that steady-state noise levels measured at each crew station within the vehicle do not exceed specified limits. The chassis operating noise level shall be measured as specified to verify adherence to requirements.

4.7.54 TOW dismount/remount test. To determine conformance to 3.10, a qualified crew shall dismount the stowed TOW components from an M901 vehicle and set-up an emplacement within 50 meters of the vehicle. The same crew shall remount the TOW components. Times required shall be measured.

5. PACKING

5.1 Preparation for delivery. Preparation for delivery and storage shall be in accordance with MIL-T-62327

6. NOTES

6.1 Intended use. The M901 is intended for, but not limited to, anti-tank combat operations. The M90l must be air transportable, and capable of crossing inland waterways under its own power.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Whether tracking and slewing rate performance shall be other than as specified (see 3.6.12.6).
c. Radio suppression, if other than as specified (see 3.6.15.5.1).
d. Painting, if other than as specified (see 3.8.1).
e. Safety markings, if other than as specified (see 3.8.3).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests and approval of the first article (see 3.1).

6.4 Inspection equipment calibration and inspection. In inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gaging, measuring, and test equipment of the required accuracy and precision. Except as otherwise specified, the contractor shall have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

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6.5 Process average. Sampling may be initiated if the process average value for the first ten vehicles inspected is less than the AQL specified in the classification of defects for major and minor defects.

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicles inspected}} \times 100$$

If the computed process average exceeds the specified AQL, 100 percent inspection shall be performed and continued until such time that the process average for ten consecutive vehicles is less than the specified AQL.

6.6 Definitions. The following definitions shall be used with the classification of defects for leaks:

a. Weep: Any evidence of fluid beyond the seal.
b. Seep: Any evidence of fluid beyond the seal that does not result in formation of a droplet.
c. Droplet: Any evidence of fluid beyond the seal that results in the formation of a droplet.
d. Drip: Any evidence of fluid beyond the seal where droplet forms and falls.
e. Leak: A steady series of drips.
f. Excessive leak: A steady stream of fluid.

6.7 Deficiency sheet. The Government inspector shall verify that a thorough inspection of each vehicle is performed by the contractor for the listed characteristics, and for any departures from good workmanship. The Government inspector shall assure that all defects encountered during inspection are enumerated on a deficiency sheet for the vehicle. Defects noted on a deficiency sheet shall contain sufficient description to enable the Government inspector and the contractor's representative to classify the defects in accordance with the classification of defects (see table IV) definitions contained in MIL-STD-105. Corrective action shall be taken for recurring defects.

6.7.1 Recurring major defects. A major defect is recurring when the same defect occurs more than once in the same sample, or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records (“P” chart or equivalent) reflect such a condition. Recurring major defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

6.7.2 Recurring minor defects. A minor defect is recurring if it occurs more than twice in the same sample or when the defect occurs in four successive samples. Recurring minor defects shall be cause for the entire lot or lots to be inspected for the recurring defects.

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6.8 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.3).

6.9 Mission failure scoring criteria. Improved TOW vehicle Failure Definition and Scoring Criteria may be obtained from US Army Tank-Automotive Command, Warren, MI 48090, ATTN: DRSTA-QRA.

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